ToyForge wireframes: read me first

Low-fidelity mobile screens (390 by 844) for the ToyForge web app, for the hackathon build. Grayscale on purpose: layout, flow, and content only. All copy is a first draft.

How to read the screens

- Gray dashed box: a placeholder for an image, 3D view, or AR view.
- Tags: Stretch means not in the hackathon core. Soon means shown but not working yet.
- Yellow dashed strip under a button: a flow or interaction note for you. It is not part of the interface.
- [Square brackets]: a value the app fills in.
- Blue button: the main action on the screen.

The flow

Onboarding (O1 to O5) leads to Home. Home has four tiles: Build, Play, Extend, Mix and match. Build is Scan, Build, Add to build, Part detail, then 3D print. Pack Rip (7, 7b) ends with a choice to 3D print, order, or skip. Play, Game Mode, and Reveal finish the loop. More holds Packs, Market, Forge Club, STEM mode, and Settings.

Behavior to keep

- Every question is a sideways scroller with the next card peeking in. The last card is always Custom (type or say it, with a mic). Only O3 shows Custom open.
- Onboarding starts with age. The options, and whether the player starts on Snap or Pro, adjust to the age. The age shows in the header after that.
- 3D print and Order stay off until the build passes the checks.
- Only 3D-printable parts can be 3D printed or ordered. Skins, glow, and trails are game only.

Not drawn yet

- Empty states (new user Toybox), scan failed, a failed check, loading while the AI works, offline.
- The order quote and delivery screens, and the Settings pages.
- The intro scene and the Reveal animation (separate brief).

What we need from design

- A look and feel for kids and parents, the logo, and an icon set.
- Pack art, character cards, and the part illustrations.
- Touch targets of 44 px or more and text contrast of 4.5 to 1.

Welcome

1 of 4

Let's make your first pack

Start with your age. We use it to pick the right options for you.

How old are you?

Swipe for more

3 4 5 6 7

Age 6

We'll start you on Snap and match the options to your age. You can change this any time.

Nickname (optional)

What should we call you?

3D PRINTER AT HOME?

Yes No Not sure

Next

Flow: your favorites

Favorites

Age 6 · 2 of 4 | Skip

Favorite toy

Pick one

- Picked — image — Cars
- image — Dolls
- image — Robots
- D

Favorite animal

Pick one

- image — Dog
- image — Cat
- Picked — image — Dragon
- S

Favorite colors

Pick up to 3

- Picked — Red
- Picked — Blue
- Green
- Y

Next

Note: Swipe sideways for more

Movies and books

Age 6 · 3 of 4 | Skip

Favorite movie

Pick one

- image [Popular pick]
- image [Popular pick]
- image [Popular pick]
- [p

Type your own | Mic

Favorite book

Pick one

- image [Popular pick]
- Picked image [Popular pick]
- image [Popular pick]
- [p

Favorite vehicle

Pick one

- image Car
- image Truck
- image Plane
- T

Next

Note: Custom opens a type or say box

A little more

Age 6 · 4 of 4 Skip

Favorite music

Picks your theme songs

- image — Pop
- image — Rock
- Picked — image — Orchestral
- E

Favorite place

Picks your skins

- Picked — image — Space
- image — Ocean
- image — Jungle
- C

When I grow up

Picks your challenges

- Picked — image — Engineer
- image — Astronaut
- image — Artist
- D

Make my pack

Flow: your first pack

Your first pack

Your first pack is ready

Designed from your picks.

Pack art made from your picks

Cars | Dragon | Red + blue | Orchestral | Space | Engineer

Inside

3 items, 1 you can 3D print

Edit my picks

Rip your first pack

Flow: Pack Rip

Home

Snap | Pro

RESUME
Red car, build 2

Continue

WHAT DO YOU WANT TO DO?



Build

Scan a toy and build upgrades for it.

Scan, builder, 3D print



Play

Bring your toy to life.

AR stage, Game Mode, Reveal



Extend

Get new parts and skills.

Market, packs, Forge Club, STEM



Mix and match

Soon

Swap skins and mash up toys.

Skins now, merge and color later

Home | Toybox | Build | Play | More

Toybox

All | Snap | Pro | 3D printed | Skins

image

Red car

Mount: top, size: small

image

Blue figure

Mount: back, size: small

image

Gear set

3D printed, Pro

image

Wing add-on

3D printed, Snap

Scan new

Flow: Scan

Home | **Toybox** | Build | Play | More

Scan

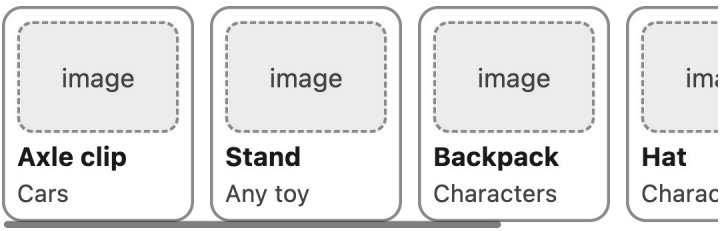
Camera view
outline overlay on the toy

Found: toy car

Attach point: top (tap to move)

Adapter

Swipe for more



Best match for the toy class comes first.

Size

Small

Large

Use seed toy

Save

Flow: Build

Home | **Toybox** | Build | Play | More

Build

Snap | Pro

3D preview
stand-in toy body at the attach point, snap grid

Undo | Redo | Delete

OK On grid | OK Fits size

OK 3D prints, no supports | OK Fits bed

Accessories | Vehicles | Characters | Effect

image	image	image	ima
Rockets 3D print	**Glow shoes** 3D print	**Wings** 3D print	**Goggl** 3D prir

AR toggle | **Pro motion test: stretch**

Preview in AR

Flow: 3D print

Home | Toybox | **Build** | Play | More

Add to build

Close

Accessories

Swipe for more

- image — **Rockets** — 3D print
- image — **Glow shoes** — 3D print
- image — **Wings** — 3D print
- ima — **Goggl** — 3D prir

Vehicles

Swipe for more

- image — **Wheels** — 3D print
- image — **Treads** — 3D print
- image — **Jet engine** — 3D print
- ima — **Prope** — 3D prir

Characters: monster, hero

Swipe for more

- image — **Monster** — 3D print
- image — **Hero** — 3D print
- image — **Robot** — 3D print
- ima — **Sideki** — 3D prir

MORE

Effects | Outfits | Gadgets | Sidekicks

Add to build

Flow: Build

Part detail

3D and AR preview of the part

Gear set, Pro

Unlocked

Stat: Speed

FITS

OK Red car, top, small

OK Blue figure, back, small

Make a skin

3D print

Equip

Add to build

Flow: Build

Home | Toybox | **Build** | Play | More

3D print

STL preview

OK On grid

OK Size class

OK No supports

OK Fits bed

3D print profile

0.4 mm nozzle, 0.2 mm layers, orientation: flat

Parts list

[number] parts, [filament] g, [time] estimate

Home

Delivery

If a check fails, tips show here and export stays off.

Export STL

Flow: Play

Pack Rip

Pack art made from your picks
tap to rip

INSIDE

? ? ?

3 items. At least 1 can be 3D printed.

Rip the pack

Flow: your items

Pack result

YOU GOT

image

Neon skin
Game only

image

Glow trail
Game only

image

Rocket boosters
3D print

WHAT NEXT FOR THE ROCKET BOOSTERS?

3D print Recommended
3D print it on your own 3D printer. The checks run first.

Order Delivered
A 3D print service makes it and delivers it. You see a quote first.

Skip
Keep it in your toybox. 3D print or order it any time.

Only the 3D-printable item can be 3D printed or ordered. Recommended comes from your printer answer in onboarding.

Play

AR stage
toy with its parts

COMPANION STATS

Speed

Spin

Glow

Replay theme song

Mini game

Game Mode

Play in AR

Flow: Mini game or Game Mode

Home | Toybox | Build | **Play** | More

Game Mode



MOOD

Joy | Suspense | Danger | Calm | Triumph | Mystery

Mic

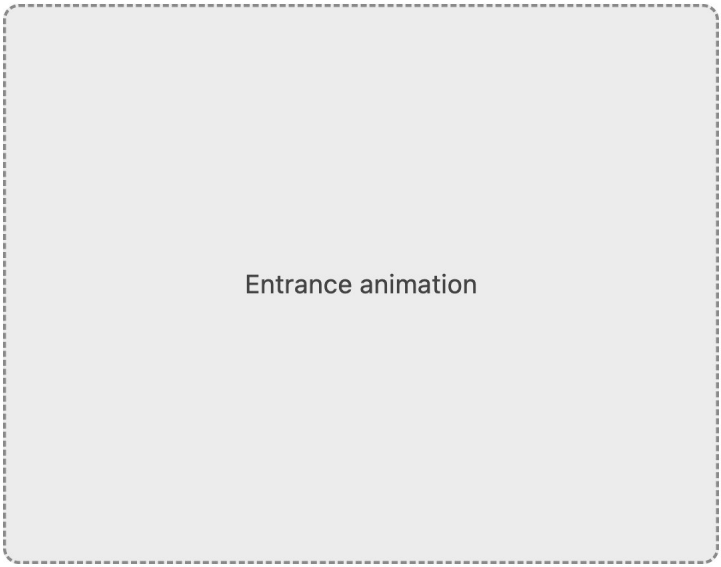
Hold to talk. Try: start adventure, danger, play my theme.

Now playing: suspense loop

Start adventure

Note: Camera scene detection comes later

Reveal



Character name

Stats and theme song title

Share

Save + Replay

Note: Plays once, then saves

More

Packs
Rip a pack, see what you own >

Market Stretch
Find parts that fit your toys >

Forge Club
Monthly sets >

STEM mode
Challenges and concept cards ☐ On

STEM challenges Stretch
Shows when STEM mode is on >

Settings
Age, line (Snap or Pro), 3D printer, nickname >

Help
How it works >

Mix and match

Merge toys Soon
Combine 2 or 3 toys

Color Soon
Paint and masks

SKINS

3D preview of your toy with the new skin

Describe a look

A neon space dragon

Mic

Space dragon | Neon | Jungle | Ice

Make skin

Flow: Part detail

Home | Toybox | Build | Play | **More**

Market

Stretch

Search

Search parts

My Toys fit | Snap | Pro | Club sets

Forge Club: Month 1

Seeded set, 6 parts, 2 skins

image

Wing add-on

Fits: Red car. 3D print: [time]

image

Gear set

Fits: Blue figure. 3D print: [time]

Open detail

Flow: Part detail

Home | Toybox | Build | Play | **More**

STEM Challenge

Stretch

Goal
Build a gear train at a set ratio

1. PREDICT

◯ Faster ◯ Slower ◯ Same

2. BUILD

Open the builder

3. TEST

Result
Measured ratio: [ ]

Concept card: gear ratio
A short explanation of the part in hand

Test it

Note: Finishing unlocks a part or skin